

Low-voltage logic spectrum

Features and functions

Increasing speed

Low Voltage Families								
	LV	LVC	AUP	AVC	ALVC	LVT	ALVT	CBTLV(D)
Supply Voltage	1 to 3.6 V*	1.2 to 3.6 V*	0.8 to 3.6 V	1 to 3.3 V	1.2 to 3.6 V	2.7 to 3.6 V	2.3 to 3.6 V	2.3 to 3.6 V
Typ. Prop. Delay	9 ns	2 ns	4 ns	2 ns	2 ns	2 ns	1.5 ns	0.15 ns
Output drive	+/- 8 mA	+/-24 mA	+/- 4 mA	+/- 8 mA	+/-24 mA	-32/64 mA	-32/64 mA	N/A**
Standby Current	20 μ A	20 μ A	0.9 μ A	20 μ A	40 μ A	120 μ A	90 μ A	10 μ A
T _{amb}	-40 °C to 125 °C	-40 °C to 125 °C	-40 °C to 125 °C	-40 °C to 85 °C	-40 °C to 85 °C	-40 °C to 85 °C	-40 °C to 125 °C	-40 °C to 125 °C
Q100 Available	Q100-Grade 1	Q100-Grade 1	Q100-Grade 1	Q100-Grade 3	Q100-Grade 3	Q100-Grade 3	Q100-Grade 1	Q100-Grade 1
Key Functions	▶ Standard logic ▶ Analog switches ▶ Timers	▶ Standard logic ▶ Analog switches ▶ Configurable and combination logic	▶ Standard logic ▶ Analog switches ▶ Configurable and combination logic	▶ Bus interface functions ▶ Level translator functions	▶ Standard logic ▶ Bus interface functions	▶ Standard logic ▶ Bus interface functions	▶ Standard logic ▶ Bus interface functions ▶ Line drivers	▶ Bus switches, ▶ Mux/DeMUX
Key Features	▶ Flow through pin out ▶ Low ground bounce and output skew	▶ 5 V tolerant I/Os ▶ Live insertion ▶ Bus hold option ▶ Termination resistor option	▶ 3.6 V tolerant I/Os ▶ Live insertion ▶ Ioff feature ▶ Low power consumption	▶ High dynamic drive ▶ Optimized for 2.5 V ▶ 3.6 V tolerant I/Os ▶ Bus hold option	▶ 5 V tolerant I/Os ▶ Bus hold option ▶ Termination resistor option	▶ 5 V tolerant I/Os ▶ Live insertion ▶ Bus hold ▶ Termination resistor option	▶ 5 V tolerant I/Os ▶ Live insertion ▶ Bus hold ▶ Termination resistor option	▶ Low R_{on} (5 Ohm typical), ▶ Internal diode for level shifting, ▶ Low prop delays
Packages	DIP, SO, SSOP, TSSOP, QFN	PicoGate, DQFN, LFBGA	PicoGate, MicroPak	PicoGate, MicroPak, TSSOP, TVSOP, QFN, BGA	SO, VSSOP, TSSOP, DQFN	SO, SSOP, BGA, TSSOP, DQFN	SSOP, TSSOP, BGA	SO, SSOP, TSSOP, DQFN, MicroPak, PicoGate,
Notes	Low voltage TTL compatible logic family	Replaces LCX logic	Replaces ULP & LVP logic	Includes level translation functions	Replaces VCX logic	Optimized for backplane applications	World's fastest standard logic family	Low voltage, fast cross bar switches
Functions Available	44	147	70	25	48	44	21	35

* Some functions can operate up to 5.5V

** Switch current is 64mA



High Voltage logic spectrum

Features and functions

Increasing speed

	High Voltage Families							
	HEF4000	HC/T	AHC/T	VHC/T	XC7	FAST	ABT	CBT(D)
Supply Voltage	5 to 15 V	2 to 6 V	2 to 5.5 V	2 to 5.5 V	2 to 5.5 V	4.5 to 5.5 V	4.5 to 5.5 V	4.5 to 5.5 V
Typ Prop Delay	60 ns	9 ns	5 ns	5 ns	5 ns	4 ns	3 ns	0.25 ns
Output Drive	+/- 3 mA	+/- 8 mA	+/- 8 mA	+/- 8 mA	+/- 8 mA	-15/24 mA	-32/64 mA	N/A*
Standby Current	600 µA	80 µA	40 µA	2 µA	2 µA	10 mA**	50 µA **	3 µA
T _{amb}	-40 °C to 85 °C (HEC: -55 °C to 125 °C)	-40 °C to 125 °C	-40 °C to 125 °C	-40 °C to 125 °C	-40 °C to 125 °C	0 °C to 70 °C (N74F Family) -40 °C to 85 °C (I74F Family)	-40 °C to 85 °C	-40 °C to 85 °C
Q100 Available	Q100-Grade 3	Q100-Grade 1	Q100-Grade 1	Q100-Grade 1	Q100-Grade 1	Q100-Grade 3***	Q100-Grade 3	Grade 3
Key Functions	- Standard logic - PLL with VCO - Counters -Timers - Comparators	- Standard logic - PLL with VCO - Analog switches - Parity generators	- Standard logic - Analog switches - Bus interface functions	- Standard logic - Line drivers	- Standard logic - Line drivers	- Standard logic - Parity generators - Bus interface functions	- Standard logic - Bus Interface functions	- Bus switches, Mux/DeMUX, - Bus exchange switches
Key Features	- High noise immunity - Over voltage tolerant I/Ps - High fanout - Extended temperature range	3 state outputs - Reduced noise - Low power - Open drain O/Ps - Extended temperature range	- Over supply voltage tolerant I/Ps - Low power consumption - Schmitt trigger	- Low power consumption - Oversupply voltage tolerant I/Ps - Schmitt trigger	- Over supply voltage tolerant I/Ps - Schmitt trigger - Low power consumption	- Termination resistor option - Flow through pin out - Open collector outputs 3 state outputs - Low output to output skew	- Bus hold option - Termination resistor option - Live insertion - Power up reset	Low R _{int} (5 Ohm typical), Internal diode for level shifting, low prop delays
Packages	DIP, SO, SSOP, TSSOP	PicoGate, MicroPak, SO, TSSOP, DQFN	PicoGate, MicroPak, SO, TSSOP, DQFN	SO, SSOP, TSSOP, QFN	TSSOP, VSSOP, QFN, MicroPak/PicoGate	DIP, SO, SSOP, TSSOP	DIP, SO, SSOP, TSSOP	SO, SSOP, QSOP, TSSOP, TVSOP, DQFN
Notes	NXP number one worldwide	Replaces older LSTTL logic	Can replace AC/T logic	Replaces VHC/T logic	Replaces TC7 logic	Replaces the Schottky diode (74s) logic	Optimized for backplane interface applications	Low impedance, extremely fast cross bar switch
Functions Available	72	311	108	10	17	80	39	20

* Switch current is 64mA

** Current for 74ABT04 and 74F04

*** I74F FAST products are Q100 Grade 3 compliant

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